

# Missing Data

Data Wrangling in R

# Dealing with Missing Data

# Missing data types

One of the most important aspects of data cleaning is missing values.

Types of “missing” data:

- **NA** - general missing data
- **NaN** - stands for “**N**ot **a** **N**umber”, happens when you do  $0/0$ .
- **Inf** and **-Inf** - Infinity, happens when you take a positive number (or negative number) by 0.

## Missing Data with Logicals

Logical operations return **NA** for **NA** values. Think about it, the data could be  $> 2$  or not we don't know, so R says there is no **TRUE** or **FALSE**, so that is missing:

```
x <- c(0, NA, 2, 3, 4)
x > 2
```

```
[1] FALSE    NA FALSE  TRUE  TRUE
```

# Missing Data Issues

Mathematical operations with **NA** result in **NAs**.

```
y <- c(1,2,3,NA)  
sum(y)
```

```
[1] NA
```

```
mean(y)
```

```
[1] NA
```

# Missing Data Issues

Logicals: TRUE is evaluated as 1 and FALSE is evaluated as 0.

```
x <- c(TRUE, TRUE, TRUE, TRUE, FALSE, NA)  
sum(x)
```

```
[1] NA
```

```
sum(x, na.rm = TRUE)
```

```
[1] 4
```

# Finding Missing data

- `is.na` - looks for `NAN` and `NA`
- `is.nan` - looks for `NAN`

```
test <- c(0, NA, -1, NaN)  
is.na(test)
```

```
[1] FALSE  TRUE FALSE  TRUE
```

```
is.nan(test)
```

```
[1] FALSE FALSE FALSE  TRUE
```

# Useful checking functions

Do we have any **NAs**? (**any** can help)

```
A <- c(1, 2, 3, NA)
B <- c(1, 2, 3, 4)
```

```
any(is.na(A)) # are there any NAs – YES/TRUE
```

```
[1] TRUE
```

```
any(is.na(B)) # are there any NAs– NO/FALSE
```

```
[1] FALSE
```



## Finding **NA** values with **count()**

Check the values for your variables, are they what you expect?

`count()` is a great option because it gives you:

1. The unique values
2. The amount of these values

Check if rare values make sense.

## Finding NA values with `count()`

```
library(readr)
library(janitor)
ufo <- read_csv(
  "https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_data_complete.csv")
```

Warning: One or more parsing issues, call ``problems()`` on your data frame for

```
dat <- vroom(...)
problems(dat)
```

```
ufo <- clean_names(ufo)

count(ufo, country)
```

```
# A tibble: 6 × 2
  country      n
  <chr>    <int>
1 au         593
2 ca        3266
3 de         112
4 gb        2050
5 us       70293
6 <NA>     12561
```

# naniar

Sometimes you need to look at lots of data... the [naniar package](#) is a good option.

```
#install.packages("naniar")  
library(naniar)
```

## naniar: `pct_complete()`

This can tell you if there are missing values in the dataset.

```
pct_complete(ufo)
```

```
[1] 97.27646
```

Or for a particular variable:

```
ufo %>% select(shape) %>%  
pct_complete()
```

```
[1] 96.4917
```

## `naniar:miss_var_summary()`

To get the percent missing (and counts) for each variable as a table, use this function.

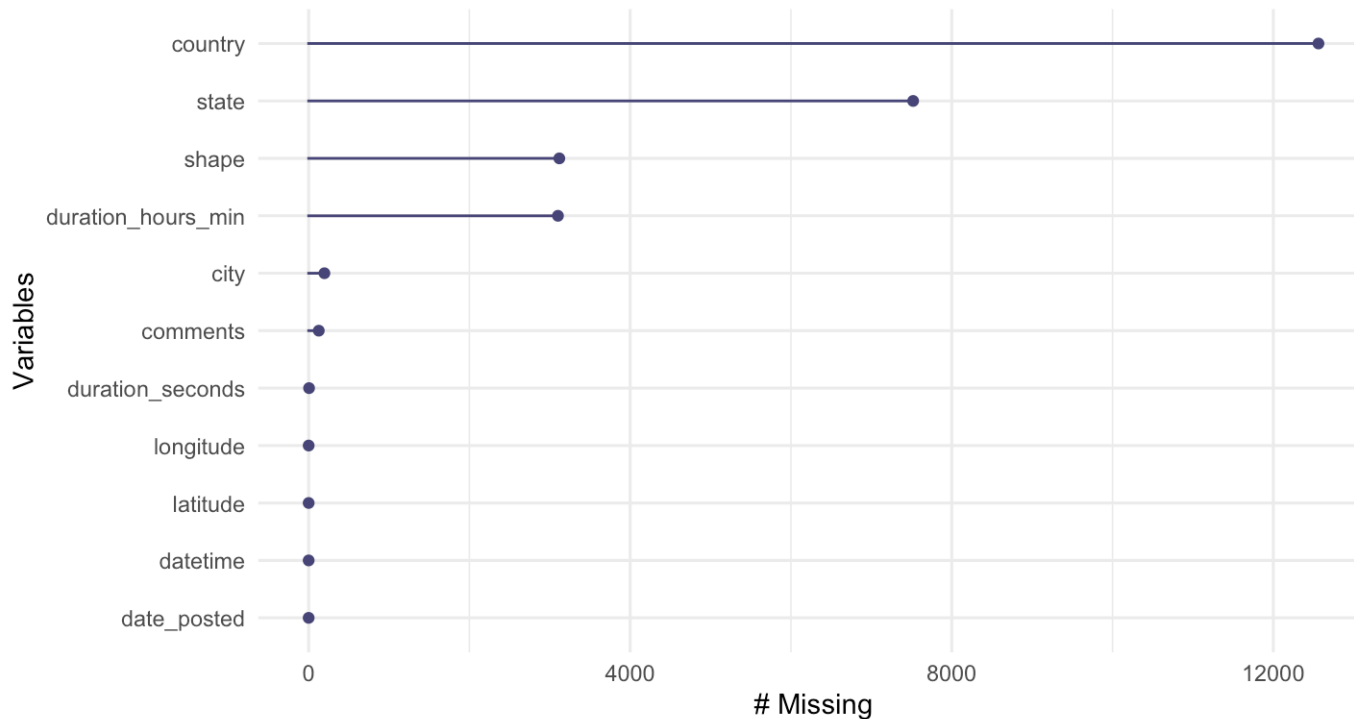
```
miss_var_summary(ufo)
```

```
# A tibble: 11 × 3
  variable      n_miss pct_miss
  <chr>      <int>    <num>
1 country    12561    14.1
2 state      7519     8.46
3 shape      3118     3.51
4 duration_hours_min 3101     3.49
5 city        196     0.221
6 comments    126     0.142
7 duration_seconds     5    0.00563
8 datetime     0     0
9 date_posted     0     0
10 latitude     0     0
11 longitude     0     0
```

## naniar plots of number missing

The `gg_miss_var()` function creates a nice plot about the number of missing values for each variable, (need a data frame).

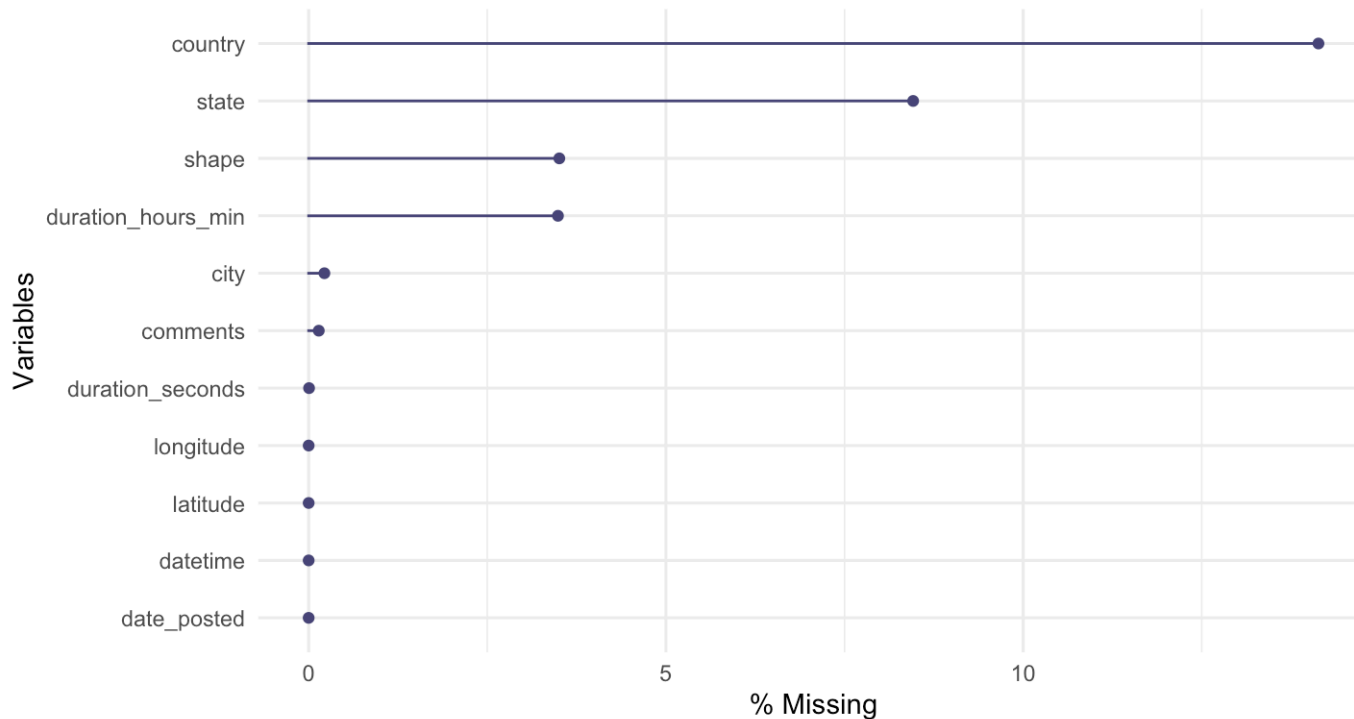
```
gg_miss_var(ufo)
```



## naniar plots of percent missing

The `gg_miss_var()` function creates a nice plot about the number of missing values for each variable, (need a data frame).

```
gg_miss_var(ufo, show_pct = TRUE)
```



## `filter()` and missing data

Be **careful** with missing data using subsetting!

**`filter()` removes missing values by default.** Because R can't tell for sure if an NA value meets the condition. To keep them need to add `is.na()` conditional.

Think about if this is OK or not - it depends on your data!



## filter() and missing data

What if NA values represent values that are so low it is undetectable?

Filter will drop them from the data.

```
count(ufo, country)
```

```
# A tibble: 6 × 2
```

|   | country | n     |
|---|---------|-------|
|   | <chr>   | <int> |
| 1 | au      | 593   |
| 2 | ca      | 3266  |
| 3 | de      | 112   |
| 4 | gb      | 2050  |
| 5 | us      | 70293 |
| 6 | <NA>    | 12561 |

```
ufo %>% filter(country == "de") %>% dim()
```

```
[1] 112  11
```

## filter() and missing data

`is.na()` can help us keep them.

```
ufo %>% filter(country == "de" | is.na(country)) %>% dim()
```

```
[1] 12673    11
```

## To remove rows with NA values for a variable use `drop_na()`

A function from the `tidyr` package. (Need a data frame to start!)

Disclaimer: Don't do this unless you have thought about if dropping NA values makes sense based on knowing what these values mean in your data.

ufo

```
# A tibble: 88,875 × 11
  datetime      city state country shape duration_seconds duration_hours_r
  <chr>         <chr> <chr> <chr>   <chr>      <dbl> <chr>
1 10/10/1949 20:... san  tx    us    cyli...    2700 45 minutes
2 10/10/1949 21:... lack... tx    <NA>    light    7200 1-2 hrs
3 10/10/1955 17:... ches... <NA>   gb    circ...     20 20 seconds
4 10/10/1956 21:... edna  tx    us    circ...     20 1/2 hour
5 10/10/1960 20:... kane... hi    us    light     900 15 minutes
6 10/10/1961 19:... bris... tn    us    sphe...     300 5 minutes
7 10/10/1965 21:... pena... <NA>   gb    circ...     180 about 3 mins
8 10/10/1965 23:... norw... ct    us    disk    1200 20 minutes
9 10/10/1966 20:... pell... al    us    disk     180 3 minutes
10 10/10/1966 21:... live... fl    us    disk     120 several minutes
# i 88,865 more rows
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,
# longitude <chr>
```

To remove rows with NA values for a variable use `drop_na()`

```
ufo %>% drop_na(state)
```

```
# A tibble: 81,356 × 11
```

|    | datetime<br><chr> |         | city<br><chr> | state<br><chr> | country<br><chr> | shape<br><chr> | duration_seconds<br><dbl> | duration_hours_r<br><chr> |
|----|-------------------|---------|---------------|----------------|------------------|----------------|---------------------------|---------------------------|
| 1  | 10/10/1949 20:... | san ... | tx            | us             | cyli...          |                | 2700                      | 45 minutes                |
| 2  | 10/10/1949 21:... | lack... | tx            | <NA>           | light            |                | 7200                      | 1-2 hrs                   |
| 3  | 10/10/1956 21:... | edna    | tx            | us             | circ...          |                | 20                        | 1/2 hour                  |
| 4  | 10/10/1960 20:... | kane... | hi            | us             | light            |                | 900                       | 15 minutes                |
| 5  | 10/10/1961 19:... | bris... | tn            | us             | sphe...          |                | 300                       | 5 minutes                 |
| 6  | 10/10/1965 23:... | norw... | ct            | us             | disk             |                | 1200                      | 20 minutes                |
| 7  | 10/10/1966 20:... | pell... | al            | us             | disk             |                | 180                       | 3 minutes                 |
| 8  | 10/10/1966 21:... | live... | fl            | us             | disk             |                | 120                       | several minutes           |
| 9  | 10/10/1968 13:... | hawt... | ca            | us             | circ...          |                | 300                       | 5 min.                    |
| 10 | 10/10/1968 19:... | brev... | nc            | us             | fire...          |                | 180                       | 3 minutes                 |

```
# i 81,346 more rows
```

```
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,
```

```
# longitude <chr>
```

## Drop all NAs with `drop_na()`

Drops rows with **any** missing data in **any** column.

```
ufo %>% drop_na() %>% dim()
```

```
[1] 69528    11
```

```
ufo %>% drop_na()
```

```
# A tibble: 69,528 × 11
```

|    | datetime   |        | city    | state | country | shape   | duration_seconds | duration_hours_r |
|----|------------|--------|---------|-------|---------|---------|------------------|------------------|
|    | <chr>      |        | <chr>   | <chr> | <chr>   | <chr>   | <dbl>            | <chr>            |
| 1  | 10/10/1949 | 20:... | san ... | tx    | us      | cyli... | 2700             | 45 minutes       |
| 2  | 10/10/1956 | 21:... | edna    | tx    | us      | circ... | 20               | 1/2 hour         |
| 3  | 10/10/1960 | 20:... | kane... | hi    | us      | light   | 900              | 15 minutes       |
| 4  | 10/10/1961 | 19:... | bris... | tn    | us      | sphe... | 300              | 5 minutes        |
| 5  | 10/10/1965 | 23:... | norw... | ct    | us      | disk    | 1200             | 20 minutes       |
| 6  | 10/10/1966 | 20:... | pell... | al    | us      | disk    | 180              | 3 minutes        |
| 7  | 10/10/1966 | 21:... | live... | fl    | us      | disk    | 120              | several minutes  |
| 8  | 10/10/1968 | 13:... | hawt... | ca    | us      | circ... | 300              | 5 min.           |
| 9  | 10/10/1968 | 19:... | brev... | nc    | us      | fire... | 180              | 3 minutes        |
| 10 | 10/10/1970 | 16:... | bell... | ny    | us      | disk    | 1800             | 30 min.          |

```
# i 69,518 more rows
```

```
# i 4 more variables: comments <chr>, date_posted <chr>, latitude <chr>,
```

```
# longitude <chr>
```

## Find columns with any missing values

Use the `miss_var_which()` function from `naniar`

```
miss_var_which(ufo) # which columns have missing values
```

```
[1] "city"           "state"          "country"  
[4] "shape"         "duration_seconds" "duration_hours_min"  
[7] "comments"
```

# Drop columns with any missing values

`miss_var_which` and function from `naniar` (need a data frame)

```
ufo %>% select(!miss_var_which(ufo))
```

```
# A tibble: 88,875 × 4
```

|    | datetime         | date_posted | latitude   | longitude    |
|----|------------------|-------------|------------|--------------|
|    | <chr>            | <chr>       | <chr>      | <chr>        |
| 1  | 10/10/1949 20:30 | 4/27/2004   | 29.8830556 | -97.9411111  |
| 2  | 10/10/1949 21:00 | 12/16/2005  | 29.38421   | -98.581082   |
| 3  | 10/10/1955 17:00 | 1/21/2008   | 53.2       | -2.916667    |
| 4  | 10/10/1956 21:00 | 1/17/2004   | 28.9783333 | -96.6458333  |
| 5  | 10/10/1960 20:00 | 1/22/2004   | 21.4180556 | -157.8036111 |
| 6  | 10/10/1961 19:00 | 4/27/2007   | 36.5950000 | -82.1888889  |
| 7  | 10/10/1965 21:00 | 2/14/2006   | 51.434722  | -3.18        |
| 8  | 10/10/1965 23:45 | 10/2/1999   | 41.1175000 | -73.4083333  |
| 9  | 10/10/1966 20:00 | 3/19/2009   | 33.5861111 | -86.2861111  |
| 10 | 10/10/1966 21:00 | 5/11/2005   | 30.2947222 | -82.9841667  |

```
# i 88,865 more rows
```

## Change a value to be NA

Let's say we think that all 0 values should be NA.

The `na_if()` function of `dplyr` can be helpful for changing all 0 values to NA. More on `mutate()` soon! (Here we save the change)

```
count(ufo, duration_seconds) %>% tail()
```

```
# A tibble: 6 × 2
  duration_seconds    n
      <dbl> <int>
1      25248000     1
2      52623200     3
3      66276000     1
4      82800000     1
5      97836000     1
6           NA     5
```



# Change a value to be NA

```
ufo <- ufo %>%  
  mutate(duration_seconds = na_if(duration_seconds, 0))
```

```
count(ufo, duration_seconds) %>% tail()
```

```
# A tibble: 6 × 2  
  duration_seconds      n  
    <dbl> <int>  
1    25248000      1  
2    52623200      3  
3    66276000      1  
4    82800000      1  
5    97836000      1  
6           NA    7228
```

## Change NA to be a value

The `replace_na()` function (part of the `tidyr` package), can do the opposite of `na_if()`. (Here we just print the change)

```
count(ufo, shape) %>% tail()
```

```
# A tibble: 6 × 2
  shape      n
  <chr>   <int>
1 round     2
2 sphere  5755
3 teardrop  817
4 triangle 8489
5 unknown  6319
6 <NA>     3118
```

## Change NA to be a value

The `replace_na()` function (part of the `tidyr` package), can do the opposite of `na_if()`. (Here we just print the change)

```
ufo %>%  
  mutate(shape = replace_na(shape, "unknown")) %>%  
  count(shape) %>% tail()
```

```
# A tibble: 6 × 2  
  shape      n  
  <chr>   <int>  
1 rectangle 1418  
2 round      2  
3 sphere    5755  
4 teardrop   817  
5 triangle  8489  
6 unknown   9437
```

# Think about **NA**

## THINK ABOUT YOUR DATA FIRST!

- ⚠ Sometimes removing **NA** values leads to distorted math - be careful!
- ⚠ Think about what your **NA** means for your data (are you sure ?).
  - Is an **NA** for values so low they could not be reported?
  - Or is it if it was too low and also if there was a different issue (like no one reported)?

## Think about **NA**

If it is something more like a zero then you might want it included in your data like a zero instead of an **NA**.

Example: - survey reports **NA** if student has never tried cigarettes - survey reports 0 if student has tried cigarettes but did not smoke that week

⚠ You might want to keep the **NA** values so that you know the original sample size.

## Word of caution

⚠ Calculating percentages will give you a different result depending on your choice to include NA values.!

This is because the denominator changes.

## Word of caution - Percentages with NA

```
count(ufo, country) %>% mutate(percent = (n/(sum(n)) *100)) %>%  
  arrange(desc(percent))
```

```
# A tibble: 6 × 3  
  country      n percent  
  <chr>    <int>   <dbl>  
1 us      70293   79.1  
2 <NA>    12561   14.1  
3 ca       3266    3.67  
4 gb       2050    2.31  
5 au        593    0.667  
6 de       112    0.126
```

## Word of caution - Percentages with NA

```
ufo %>% drop_na(country) %>%  
count(country) %>% mutate(percent = (n/(sum(n)) *100)) %>%  
  arrange(desc(percent))
```

```
# A tibble: 5 × 3  
  country      n percent  
  <chr>    <int>   <dbl>  
1 us      70293   92.1  
2 ca      3266    4.28  
3 gb      2050    2.69  
4 au       593    0.777  
5 de       112    0.147
```

Should you be dividing by the total count with **NA** values included?  
It depends on your data and what **NA** might mean.  
Pay attention to your data and your **NA** values!



# Summary

- `is.na()`, `any(is.na())`, `count()`, and functions from `naniar` like `gg_miss_var()` can help determine if we have **NA** values
- `filter()` automatically removes **NA** values - can't confirm or deny if condition is met (need `| is.na()` to keep them)
- `drop_na()` can help you remove **NA** values from a variable or an entire data frame
- **NA** values can change your calculation results
- think about what **NA** values represent - don't drop them if you shouldn't

[Link to Lab](#)